

Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science

Expected May 2028

- GPA: 3.9 | SAT: 1590

- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia

Aug 2025 – Present

- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.
- Pipeline processes **1000fps** pile-driving footage — YOLO detection and segmentation to track painted depth markings, audio-visual sync to isolate hammer strikes, and depth-map pixel-to-real-world conversion.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly informing model design decisions**.

Undergraduate Researcher, Abbasi Lab, UCSF

Sep 2026 – Present

- Applying **unsupervised and semi-supervised deep learning** (deep autoencoders, non-negative matrix factorization, semi-supervised CNNs) to MRI/fMRI from thousands of UCSF patients and Allen Institute gene expression images.

Software Engineering Intern, CACI International Inc. – Virginia

May 2025 – Aug 2025

- Built a full-stack logistics web app for **U.S. Customs field officers** using Angular, Spring Boot, and PostgreSQL.
- Expanded database test coverage **by 20%**, enhancing system stability and enabling more efficient bug detection.
- Ramped onto a large, unfamiliar Spring Boot codebase fast enough to **pick up frontend bug tickets and a backend bug-fix story**, contributing through sprint planning, standups, and code reviews in an Agile team.

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA

Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using Levenshtein distance over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus (**~1M annotated words across ~10K texts**).
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

Projects

PotBot — Civic Reporting Agent, CalHacks Prize Winner

<https://devpost.com/software/potbot>

- Won sponsor prize at CalHacks — led team of 3, owned backend and CV pipeline, building an agentic pipeline with a Llama vision model to validate infrastructure damage photos and auto-submit repair requests to SF city government via Playwright; **shipped working SF demo in 36 hours** over an unfinished feature set.

DealScout — Resale Arbitrage Agent (LangGraph, AWS)

github.com/FavouritePlayer/dealscout-cloud

- Migrated a hackathon prototype into a 4-service system — containerized with Docker, deployed via Kubernetes (k3s) on AWS EC2, provisioned with Terraform — **keeping a full multi-hour AWS demo under \$0.10**.

Context Custodian — Workspace Hygiene Agent (FastAPI)

github.com/FavouritePlayer/ContextCustodian

- Built a FastAPI agent that compacts and cleans an agent-first workspace for downstream agents to ingest — a three-stage prompt-injection detection pipeline (regex, vector similarity, LLM adjudication) with a fail-open fallback, where **every fix is a reversible move, never a delete**.

Quadruped Locomotion with PPO (MuJoCo, Gymnasium, SB3)

github.com/FavouritePlayer/ant_sim

- Trained and validated quadruped locomotion policies (PPO, MuJoCo) through an **automated regression-test suite and multi-seed replication**, shipping a policy with 2.1x the flat-baseline reward on unseen terrain.

Face Morpher — From-Scratch Warping (NumPy, dlib, scipy)

github.com/FavouritePlayer/face-morpher

- Replaced OpenCV's warp with a **custom NumPy warp** (affine solve, inverse mapping, bilinear interpolation).

Skills

Languages: Python, Java, JavaScript, SQL, Dart, Scheme

Technologies: React.js, Angular, Spring Boot, Flutter, Next.js, Firebase, AWS, Docker, Kubernetes, Terraform, MongoDB, Git, PyTorch, scikit-learn, Pandas, LangGraph, FastAPI, Playwright, MuJoCo, Gymnasium, Stable-Baselines3.